

Heat Transfer

PRE-LAB PLAN

Unit Operations Lab 4

2/3/2026

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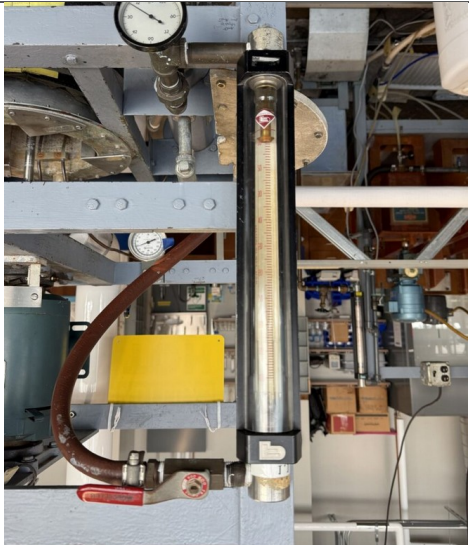
1. Experimental Objective

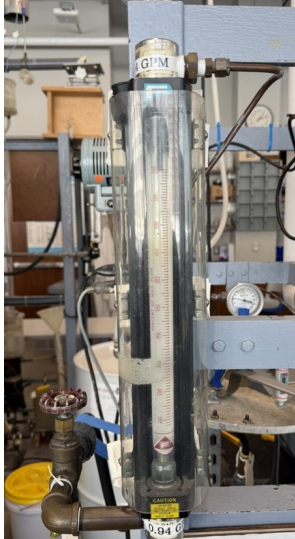
The purpose of the Heat Transfer lab is to experimentally determine the effect impeller speed, water flow rate through the heat exchanger and vessel temperature on the Q_{13} , heat transfer coefficient H_i , and on the overall heat transfer coefficient U . This is explored in two experiments with the tank initially undisturbed and then later being baffled. This is accomplished by varying the steam flowrate, cooling water flow rate and recirculation flow rate and examining how it affects the temperature.

2. Experimental

This lab is divided into two experiments: Experiment 1 involves heat transfer within a circulated tank without the presence of baffles and Experiment 2 follows an identical procedure as the first one while having baffles present within the tank. Using a stirred and jacketed vessel, the main goal of both experiments is to calculate the heat transfer coefficients at various conditions. A summary of the apparatus is provided below.

2.1 Apparatus

<i>Number</i>	<i>Name</i>	<i>Main vs Auxiliary</i>	<i>Description</i>	<i>Image</i>
1	H.E. Flowmeter	Auxiliary	Reads flow (of cooling water) from Heat Exchanger as % of 1.1 GPM	

2	Coil Flowmeter	Auxiliary	Reads flow (of cooling water) into coil as % of 0.94 GPM	
3	Tank	Main	Vessel that houses: coil, propellor & baffles (in part 2)	
4	Propellor motor	Main	Unit that powers rotation of propellor in Tank	
5	Coil	Main	Carries cooling water throughout tank	

6	Water Heater	Main	Heats water	
7	H.E. Rotameter	Auxiliary	Controls water flow in heat exchanger	
8	Controller Unit (Propellor)	Main	Controls the output to the Power Unit, thus controlling rotation of the propellor	
9	Pump	Main	Pushes water throughout the system	
10	Temperature Gauges (1-7)	Auxiliary	(In order from top left going right) T1. Takes temp. reading of water into Coil T2. Takes temp. reading of	

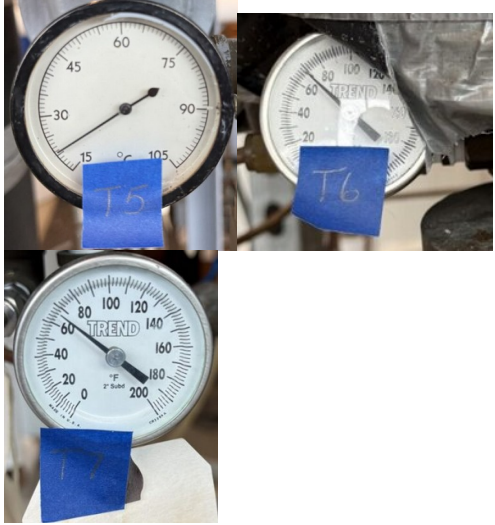
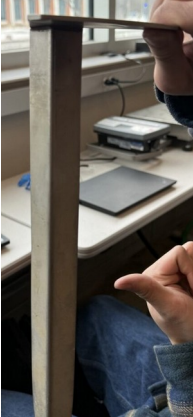
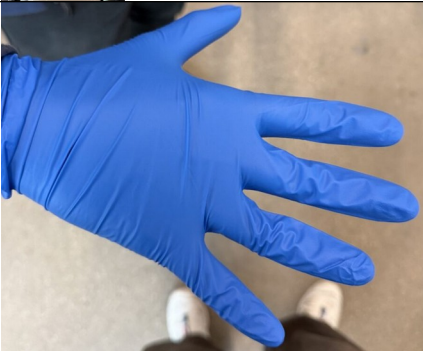
			Tank T3. Takes temp. reading coming out of Coil T4. Takes temp. reading of water from H.E. T5. Takes temp reading of water to H.E. T6. Takes temp. reading of water to condenser T7. Takes temp. reading out of condensor	
11	Condenser	Main	Used to condense steam from tank	
12	Tachometer	Auxiliary	Display the rpm of propeller	

Table 1: Summary of the Apparatus

2.2 Material, Supplies & Additional Equipment

Materials/ Supplies	Material or supply	Description	Image
Baffle	Supply	Used to interrupt/change flow of steam in Tank	
Steam	Material	Used in experiment for heat exchange	
Glove	Supply	To protect and cover hands	
Safety Goggles	Supply	To cover eyes and reduce chance of contaminants coming in contact with eyes	

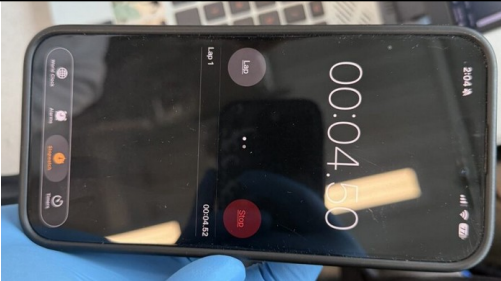
Timer	Supply	Used to track time in between readings	
Measuring Cylinder	Supply	Used to measure condensate	
Heat resistant gloves	Supply	Prevent burns from hot surfaces and safe handling of measuring cylinder	

Table 2 : Summary of Materials and Supplies

2.3 Experimental Plan

Experiment 1: Unbaffled Stirred Tank

Experimental Variables:

Independent Variables: Impeller Speed, Recirculation Flowrate, and Cooling coil flowrate

Dependent Variables: Temperature from all temperature gauges, Steam condensate rate, Flow rates of the cooling coil and recirculation flow

Experimental Setup:

1. Fill the tank with water to a level where it is about 2 inches from the top of the tank. Then proceed to put the lid on the tank
2. Turn on the impeller and adjust the speed to about 300 RPM. The proppeler will be turning counterclockwise.
 - a. Do note that the RPM fluctuates heavily between 5-25 RPM
3. Turn on the cooling water with Valve 1 and 2. From there open the valve to get 50% flowrate on the rotameter
4. Turn on the cooling water to the tube side of the heat exchanger V11 and also adjust the flowmeter to about 0.5 LPM
5. Open the condenser valve 5 to let cooling water flow through the stream condenser
6. Proceed to prime the pump. Have the outlet valve 9 closed and Valve 12 and 6 ope. Then switch on the pump and let it run for a few seconds to remove air from the pump head and piping
7. Switching the recirculation flow. Open Valve V4, open V9 slowly to 100% flow and then close V6. Keep V12 open. Let the water flow 100% flow rate to purge air in the line then reduce it back to 50%.

Experimental Procedure:

1. Open steam valve 3 slightly to about $\frac{1}{2}$ to 1 turn. This lets the steam enter the jacket and condenses completely.
2. Set the impeller speed to the first value. About 300 RPM and start taking periodic measurements of the temperaturs and flowrates. This is to determine if the temperatures have stabilized.
3. Adjust the cooling coil water flowrates or the recirculation flow that's in the external heat exchanger. Try to keep the vessel temperature between 65C and 85C
4. Once steady state is reached record the following data
 - a. Flow rates of the cooling coil and recirculation flow
 - b. Temperatures of all the temperature gauges
 - c. Amount of steam condensate collected.
5. Repeat the steps above for two more impeller speeds.
6. Adjust the recirculation water flow rate again to maintain temperature and ensure that steady state is reached before taking data.

Experiment 2: Baffled Stirred Tank

Experimental Variables:

Independent Variables: Baffles, Impeller Speed, Recirculation Flowrate, and Cooling coil flowrate

Dependent Variables: Temperature from all temperature gauges, Steam condensate rate, Flow rates of the cooling coil and recirculation flow

Experimental Procedure:

1. Conduct the same set of experiments as experiment 1 except with the 4 baffles into the vessel
2. You are able to conduct this experiment alongside the previous section.
3. To do this, turn the pump off and carefully remove the vessel cover plate and slip the baffles into place.
 - a. Make sure the top flat section is seated properly and secured to the allen screw heads.
4. Turn the pump back on.
5. Repeat steps from the previous experiment.

Data Analysis

You will want to calculate the following:

- Overall Heat Transfer Coefficient
- Outer Heat Transfer coefficient
- Heat transfer Coefficient
- Heat Transfer Rate

3. Project Safety Evaluation

Hazard	Yes/No	If Yes, Description
Potential electrical hazards?	yes	Pump and agitator are powered by electricity and the system is full of water so if any water makes it onto electrical components a shock hazard persists
Potential mechanical hazards?	yes	Rotating assembly for the pump and rotating agitator shaft pose a mechanical hazard
Potential pressure hazards?	no	

Potential temperature hazards?	yes	System will be supplied with low pressure steam at approximately 100C and the main process fluid will be +60C
Hazardous/toxic chemicals involved	No	
Gloves required?	Yes	Gloves are always required for general lab safety purposes regardless of perceived hazard. Heat resistant gloves are also necessary for operating the unit as many surfaces are +60C.
MSDS reviewed by all team members	Yes	Water (7732-18-5)
Process parameters	Max Expected	List location and possible hazards and precautions
Temperature (°C)	100	
Pressure (psia)	30	
Safety controls	none	

I have read relevant background material for the Unit Operations Laboratory entitled: “Process Control: Level Control” and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

Alexis Guillen	2/3/26
Signature of team member	Date
Khoa Nguyen	2/3/26
Signature of team member	Date
Rishabh S. Patel	2/3/26
Signature of team member	Date
Kyle Hankosky	2/3/26
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